

CB23731-DATA VISUALIZATION TECHNIQUES

LAB MANUAL

EXPERIMENT 1A

SETTING UP PYTHON ENVIRONMENT AND LIBRARIES – JUPYTER NOTEBOOK

Aim

To install Python, configure Jupyter Notebook, understand the notebook environment, execute Python programs, create Markdown cells, and use basic Jupyter Notebook features for Data Analytics applications.

Algorithm

Algorithm 1: Installing Python

Step 1: Download Python from python.org

Step 2: Run the installer

Step 3: Select "Add Python to PATH"

Step 4: Click Install Now

Step 5: Verify installation using:

`python --version`

Step 6: Observe installed Python version.

Algorithm 2: Installing Jupyter Notebook

Step 1: Open Command Prompt

Step 2: Install Jupyter

`pip install notebook`

Step 3: Launch Jupyter Notebook

`jupyter notebook`

Step 4: Browser automatically opens Notebook Dashboard.

Algorithm 3: Creating a Notebook

Step 1: Open Jupyter Dashboard

Step 2: Select New → Python 3

Step 3: Create Notebook

Step 4: Rename Notebook

Step 5: Create Code and Markdown Cells

Step 6 : Execute cells using Shift + Enter

Result:

Python and Jupyter Notebook has been installed successfully.

EXPERIMENT 1B

DATA MINING FRAMEWORK

Aim

To install and import libraries, import dataset and perform data pre-preprocessing using CRISP framework.

Algorithm

Algorithm 1: Installing Python

Step 1: Import pandas to read dataset.

Step 2: Read the dataset file

Step 3: Find the first few rows and the number of features.

Step 4: Perform data pre-processing and get the total stroke output.

Step 5: Visualize the data using matplotlib library.

Program:

```
In [11]: import pandas as pd

df = pd.read_csv(r"C:\Users\hp\Downloads\healthcare-dataset-stroke-data.csv")

print(df.head())
```

	id	gender	age	hypertension	heart_disease	ever_married	\
0	9046	Male	67.0	0	1	Yes	
1	51676	Female	61.0	0	0	Yes	
2	31112	Male	80.0	0	1	Yes	
3	60182	Female	49.0	0	0	Yes	
4	1665	Female	79.0	1	0	Yes	

	work_type	Residence_type	avg_glucose_level	bmi	smoking_status	\
0	Private	Urban	228.69	36.6	formerly smoked	
1	Self-employed	Rural	202.21	NaN	never smoked	
2	Private	Rural	105.92	32.5	never smoked	
3	Private	Urban	171.23	34.4	smokes	
4	Self-employed	Rural	174.12	24.0	never smoked	

	stroke
0	1
1	1
2	1
3	1
4	1

```
In [13]: print(df.info())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5110 entries, 0 to 5109
Data columns (total 12 columns):
#   Column                Non-Null Count  Dtype  
---  -
0   id                     5110 non-null   int64  
1   gender                 5110 non-null   object  
2   age                   5110 non-null   float64 
3   hypertension           5110 non-null   int64  
4   heart_disease          5110 non-null   int64  
5   ever_married           5110 non-null   object  
6   work_type              5110 non-null   object  
7   Residence_type         5110 non-null   object  
8   avg_glucose_level      5110 non-null   float64 
9   bmi                   4909 non-null   float64 
10  smoking_status         5110 non-null   object  
11  stroke                 5110 non-null   int64  
dtypes: float64(3), int64(4), object(5)
memory usage: 479.2+ KB
None
```

```
In [14]: print(df.describe())
```

	id	age	hypertension	heart_disease \
count	5110.000000	5110.000000	5110.000000	5110.000000
mean	36517.829354	43.226614	0.097456	0.054012
std	21161.721625	22.612647	0.296607	0.226063
min	67.000000	0.080000	0.000000	0.000000
25%	17741.250000	25.000000	0.000000	0.000000
50%	36932.000000	45.000000	0.000000	0.000000
75%	54682.000000	61.000000	0.000000	0.000000
max	72940.000000	82.000000	1.000000	1.000000

	avg_glucose_level	bmi	stroke
count	5110.000000	4909.000000	5110.000000
mean	106.147677	28.893237	0.048728
std	45.283560	7.854067	0.215320
min	55.120000	10.300000	0.000000
25%	77.245000	23.500000	0.000000
50%	91.885000	28.100000	0.000000
75%	114.090000	33.100000	0.000000
max	271.740000	97.600000	1.000000

```
In [6]: print(df.isnull().sum())
```

```
id                0
gender            0
age              0
hypertension      0
heart_disease     0
ever_married      0
work_type         0
Residence_type    0
avg_glucose_level 0
bmi              0
smoking_status    0
stroke            0
dtype: int64
```

```
: print(df["stroke"].value_counts())
```

```
stroke
0    4861
1     249
Name: count, dtype: int64
```

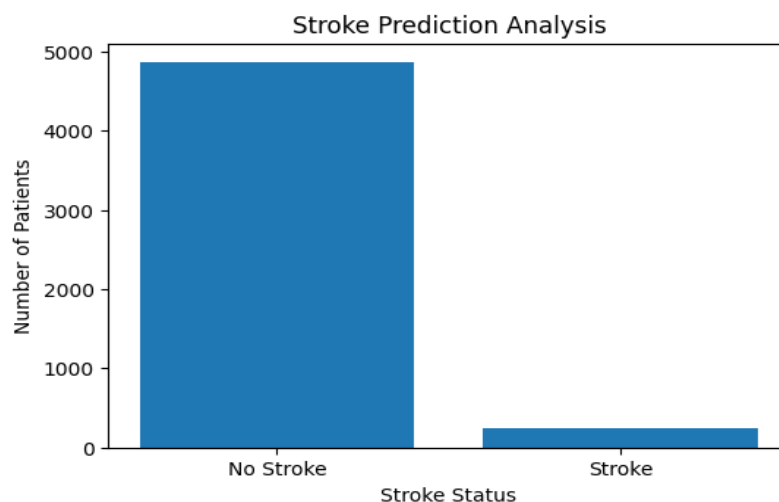
```
In [15]: import matplotlib.pyplot as plt

stroke_counts = df["stroke"].value_counts()

plt.figure(figsize=(6,4))
plt.bar(["No Stroke", "Stroke"], stroke_counts)

plt.title("Stroke Prediction Analysis")
plt.xlabel("Stroke Status")
plt.ylabel("Number of Patients")

plt.show()
```



Result:

The heart_stroke dataset has been loaded, pre-processed, visualized successfully using CRISP framework.